

Lost & Found System with Automatic Matching

A PROJECT REPORT

For

**Major Project (CA301P)
Session (2025-26)**

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Submitted in partial fulfilment of the Requirements for the Degree of

MASTER OF COMPUTER APPLICATIONS

**Under the Supervision of
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Submitted to

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CERTIFICATE

Certified that **ARUN KUMAR (202410116100040), AMAN KUMAR (202410116100020), ANURAG SINGH KUSHWAHA (202410116100039), ANTRA PRAKASH (202410116100035)** have carried out the project work having “**Lost & Found System with Automatic Matching**” (CA301P) for **Master of Computer Applications** from KIET Group of Institutions (an Deemed to be University) under Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow under my supervision. The project report embodies original work, and studies are carried out by the student himself/herself and the contents of the project report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

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ABSTRACT

In today's modern and fast-moving environment, people frequently lose valuable personal belongings such as wallets, mobile phones, ID cards, keys, bags, documents, and electronic devices. Traditional methods of managing lost and found items rely on manual registers, help desks, social media posts, and notice boards, which are inefficient, decentralized, and time-consuming. These methods often fail to establish a proper connection between the owner of the lost item and the person who found it, resulting in a low recovery rate.

The **Lost & Found System with Automatic Matching** is a web-based application developed to provide a centralized and intelligent platform for reporting and recovering lost items. The system enables users to register, log in, and report lost or found items with details such as item name, description, date, location, and optional images. All data is stored securely in a centralized database.

The major feature of the system is its automatic matching mechanism. The application uses intelligent matching techniques such as keyword comparison, TF-IDF (Term Frequency–Inverse Document Frequency), and cosine similarity to compare descriptions of lost and found items. Based on the similarity score, the system automatically suggests possible matches and notifies users.

The proposed system is developed using modern web technologies including HTML, CSS, JavaScript, Flask and MySQL. The platform is scalable, secure, user-friendly, and suitable for implementation in colleges, offices, airports, shopping malls, railway stations, and other public spaces.

The system improves efficiency, minimizes manual effort, enhances recovery chances, and contributes toward smart digital infrastructure and resource management.

Keywords: Lost & Found System, Automatic Matching, NLP, TF-IDF, Cosine Similarity, Flask, Web Application, Information Retrieval, Digital Infrastructure.

ACKNOWLEDGEMENT

We express our sincere gratitude to our project supervisor, **Mr. Roshan Kumar Gupta**, Assistant Professor, Department of Computer Applications, KIET Group of Institutions, for his continuous guidance, valuable suggestions, encouragement, and support throughout the development of this project.

We also extend our heartfelt thanks to the faculty members of the MCA Department for providing us with the necessary resources and academic support required for successful completion of this project.

We are grateful to our friends and classmates for their motivation and cooperation during different phases of the project work. Finally, we sincerely thank our parents and family members for their constant encouragement, support, and blessings.

Without their support and motivation, the completion of this project would not have been possible.

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INTRODUCTION

1.1 Overview

In the modern digital era, people frequently lose valuable belongings such as mobile phones, wallets, ID cards, keys, bags, and important documents in public and institutional environments. Places such as colleges, offices, airports, railway stations, shopping malls, and corporate campuses witness thousands of lost item cases every year. Recovering these belongings becomes difficult because traditional lost and found systems are mostly manual and unorganized.

Traditional methods include:

- Notice boards
- Handwritten registers
- Social media announcements
- Verbal communication

These methods suffer from several limitations such as lack of centralized storage, manual searching, poor organization, and low recovery success rates. There is no intelligent mechanism available to automatically compare lost and found reports.

To solve this problem, the Lost & Found System with Automatic Matching is proposed. The system is a web-based application that allows users to report lost or found items digitally. All reports are stored in a centralized database, and intelligent matching algorithms compare item descriptions to identify possible matches automatically.

The system integrates web technologies and Natural Language Processing (NLP) techniques such as:

- Keyword Matching
- TF-IDF
- Cosine Similarity

These techniques improve matching accuracy even when item descriptions are written differently.

Example:

Lost Item Description:

“Black Samsung phone with cracked screen”

Found Item Description:

“Samsung mobile, black color, broken display”

The system intelligently identifies these descriptions as similar and generates a possible match.

The proposed platform improves item recovery efficiency, reduces manual effort, and provides a secure and scalable digital solution.

1.2 Objectives

The main objectives of the Lost & Found System are:

1. Centralized Platform

To develop a centralized web-based platform for reporting lost and found items.

2. Automated Matching

To automate the matching process using NLP-based similarity algorithms.

3. Faster Recovery

To reduce the time required to recover lost belongings.

4. Improved Accuracy

To improve matching accuracy using intelligent text comparison methods.

5. Notification System

To notify users whenever a possible match is identified.

6. Secure Communication

To provide secure communication between the owner and finder.

7. Fraud Prevention

To prevent fake claims through administrative verification.

8. Scalable Architecture

To design a scalable system suitable for public and institutional use.

1.3 Importance of AI

Artificial Intelligence plays a significant role in improving the efficiency and intelligence of the Lost & Found System. Traditional systems rely heavily on manual comparison of reports, which is time-consuming and inaccurate.

AI-based Natural Language Processing techniques help the system understand and compare textual descriptions intelligently.

AI Techniques Used

Keyword Matching

Important keywords are extracted and compared.

TF-IDF

TF-IDF measures the importance of words in item descriptions.

Cosine Similarity

Cosine similarity calculates similarity between text vectors.

Benefits of AI Integration

- Faster matching process
- Reduced manual effort
- Improved recovery accuracy
- Intelligent text understanding
- Better user experience

The integration of AI transforms the system from a simple database platform into an intelligent automated recovery solution.

1.4 Scope

The scope of the Lost & Found System extends across multiple domains and environments.

The system can be implemented in:

- Educational institutions
- Corporate offices
- Airports
- Railway stations
- Shopping malls
- Public transport systems
- Event venues

The project can further be expanded with features such as:

- Mobile application support
- AI-based image recognition
- GPS-based location matching
- Chat functionality
- QR-code integration

The system architecture is scalable and flexible for future enhancements.

1.5 Limitations

Although the system provides intelligent automation, it also has certain limitations:

1. Dependence on Description Quality

Matching accuracy depends on how clearly users describe items.

2. Internet Dependency

The system requires internet connectivity for operation.

3. Limited Image Processing

Advanced image recognition is not included in the current version.

4. Human Error

Users may provide incorrect or incomplete information.

Despite these limitations, the system significantly improves over traditional methods.

1.6 Motivation

The motivation behind developing the Lost & Found System arises from the common real-world problem of losing personal belongings.

Existing systems are inefficient, slow, and heavily dependent on manual effort. Many people fail to recover lost belongings because there is no structured mechanism for matching reports.

The project aims to solve this problem using intelligent digital technologies by integrating:

- Web Development
- Database Management
- Natural Language Processing
- Information Retrieval Techniques

The system demonstrates how intelligent automation can solve practical everyday problems efficiently.

1.7 Hardware and Software Requirements

Hardware Requirements

Component	Requirement
Processor	Intel i3 or above
RAM	Minimum 4 GB
Storage	20 GB Free Space
Internet	Required
Camera	Required

Software Requirements

Frontend Technologies

- HTML
- CSS
- JavaScript
- Bootstrap

Backend Technologies

- Flask

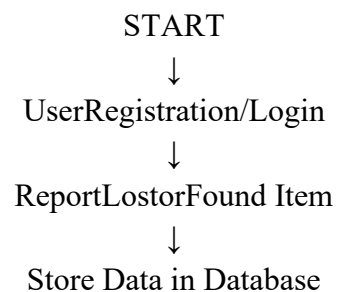
Database

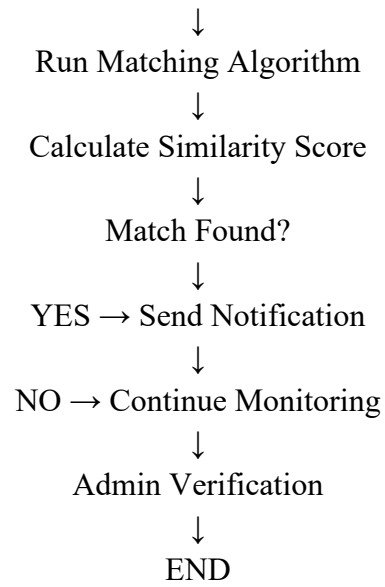
- MySQL

Development Tools

- Visual Studio Code
- GitHub
- Chrome Browser

1.8 Flowchart





LITERATURE REVIEW

2.1 Introduction

Lost and found management has been an important issue in educational institutions, offices, airports, and public places for many years. Traditional systems mainly depend on manual methods such as handwritten registers, notice boards, and verbal communication. These methods are slow, unorganized, and inefficient.

With the growth of digital technologies, some organizations introduced web-based lost and found portals. However, most of these systems only store records and do not provide intelligent matching between lost and found items.

The proposed Lost & Found System improves this process by using Natural Language Processing (NLP) techniques and intelligent matching algorithms.

2.2 Evolution and Existing Approaches

Lost and found systems evolved from manual methods to digital platforms.

Manual Systems

Traditional systems used registers and notice boards for maintaining records. These methods were time-consuming and difficult to manage.

Social Media Platforms

Users started posting lost item details on social media platforms such as WhatsApp and Facebook. Although communication became faster, there was no proper organization or verification.

Web-Based Systems

Some institutions developed online portals for submitting lost and found reports. These systems improved record storage but still lacked automatic matching functionality.

AI-Based Systems

Modern systems use Artificial Intelligence and NLP techniques such as keyword matching, TF-IDF, and cosine similarity to compare descriptions automatically and identify possible matches.

2.3 Limitations of Existing Systems

Existing systems suffer from several limitations:

- No centralized database
- Manual comparison of records
- Time-consuming process
- Low recovery success rate
- Lack of intelligent matching
- Security and verification issues

These limitations reduce the efficiency of traditional lost and found systems.

2.4 Comparative Analysis and AI Technology

System Type	Limitation
Manual Registers	Slow and unorganized
Notice Boards	Difficult searching
Social Media	No verification
Basic Web Portals	No automatic matching
AI-Based Systems	Better and more efficient

The proposed system uses AI-based techniques including:

Keyword Matching

Compares important words in item descriptions.

TF-IDF

Measures word importance in text data.

Cosine Similarity

Calculates similarity between descriptions.

These techniques improve matching accuracy and recovery rates.

2.5 Research Gap

Most existing systems still rely heavily on manual searching and lack intelligent automation. There is a need for a centralized platform that can automatically compare descriptions and notify users about possible matches.

The proposed Lost & Found System addresses this gap by integrating:

- Centralized database management
- NLP-based matching algorithms
- Notification mechanisms
- User authentication and admin verification

2.6 Importance and Summary

Intelligent lost and found systems improve recovery efficiency by reducing manual effort and automating the matching process. AI-based text similarity techniques help identify matches even when descriptions are written differently.

The literature review shows that integrating web technologies, databases, and NLP algorithms provides a more efficient and reliable solution compared to traditional methods.

SYSTEM ANALYSIS

3.1 Introduction

System analysis is the process of studying and understanding the requirements, functionality, limitations, and overall working of a system before implementation. In the Lost & Found System, system analysis helps identify the problems in traditional lost item management methods and provides an intelligent digital solution to overcome those issues.

The proposed system focuses on automating the process of reporting and matching lost and found items using web technologies and Natural Language Processing techniques. The analysis phase helps in understanding user requirements, system behaviour, and operational feasibility.

3.2 Limitations of Existing System

Traditional lost and found systems suffer from several limitations that reduce efficiency and recovery success rates.

Manual Processes

Most organizations maintain handwritten registers or notice boards, which require manual searching and comparison.

Lack of Centralization

Records are scattered across different departments or locations, making searching difficult.

No Automatic Matching

Existing systems do not provide intelligent matching between lost and found descriptions.

Time-Consuming

Users often spend days or weeks searching for lost belongings.

Security Issues

Fake claims and unauthorized access are common problems.

Poor Organization

Large amounts of data become difficult to manage manually.

These limitations highlight the need for an automated and centralized solution.

3.3 Proposed System

The proposed Lost & Found System provides a smart and centralized platform for managing lost and found records digitally.

The system allows users to:

- Register and log in
- Report lost items
- Report found items
- View possible matches
- Receive notifications

The system stores all information in a centralized database and automatically compares descriptions using intelligent matching algorithms.

Main features of the proposed system include:

Automatic Matching

The system compares descriptions using:

- Keyword Matching
- TF-IDF
- Cosine Similarity

Centralized Database

All records are stored securely in one platform.

Notification System

Users receive notifications when a possible match is found.

Admin Verification

Administrators can monitor reports and prevent fake claims.

The proposed system improves efficiency, reduces manual effort, and increases recovery success rates.

3.4 Objectives of the Proposed System

The major objectives of the system are:

- To automate the matching process for lost and found items.
- To provide a centralized digital platform.

- To improve recovery rates using intelligent algorithms.
- To reduce time and manual effort.
- To provide secure communication between users.
- To prevent fraudulent claims through admin verification.
- To create a scalable and user-friendly system.

3.5 Feasibility Study

Feasibility analysis determines whether the proposed system can be implemented successfully.

Technical Feasibility

The system uses technologies such as:

- HTML
- CSS
- JavaScript
- MySQL

These technologies are easily available and suitable for implementation.

Economic Feasibility

The project requires minimal hardware and software resources, making it cost-effective.

Operational Feasibility

The system is easy to use and can be implemented in colleges, offices, airports, and public places without major operational difficulties.

3.6 Functional Requirements

The functional requirements of the system are:

- User Registration
- User Login
- Report Lost Item

- Report Found Item
- Automatic Matching
- Notification System
- Admin Dashboard
- Database Management

3.7 Non-Functional Requirements

The non-functional requirements include:

Security

User information must remain protected.

Reliability

The system should provide accurate matching results.

Scalability

The platform should support increasing users and records.

Performance

The system should generate matches quickly.

Usability

The interface should be simple and user-friendly.

3.8 Constraints of the System

The system also has certain constraints:

- Matching accuracy depends on description quality.
- Internet connectivity is required.
- Advanced image recognition is not included in the current version.
- Incorrect user inputs may affect results.

Despite these constraints, the proposed system performs significantly better than traditional methods.

3.9 Summary

The system analysis phase identified the major limitations of existing lost and found systems and proposed an intelligent automated solution. The Lost & Found System integrates centralized database management, intelligent matching algorithms, and notification mechanisms to improve item recovery efficiency.

The analysis confirms that the system is technically feasible, economically affordable, operationally practical, and suitable for real-world implementation.

SYSTEM DESIGN

4.1 Introduction

System design is the process of converting system requirements into a structured technical solution. It defines how different components of the system interact with each other and how data flows throughout the application.

The Lost & Found System is designed as a web-based platform that provides centralized management of lost and found records along with intelligent automatic matching. The design focuses on scalability, efficiency, security, and user-friendliness.

The system consists of three major layers:

- Presentation Layer
- Application Layer
- Data Layer

These layers work together to provide smooth communication between users, backend services, and the database.

4.2 System Architecture

The Lost & Found System follows a three-tier architecture.

1. Presentation Layer

This layer contains the frontend user interface where users interact with the system.

Functions:

- User Registration
- User Login
- Report Lost Item
- Report Found Item
- View Matches
- Notification Display

Technologies Used:

- HTML
- CSS
- JavaScript
- Bootstrap

2. Application Layer

This layer handles business logic and matching algorithms.

Functions:

- Authentication
- Data Processing
- Matching Logic
- Notification Generation
- Admin Verification

Technologies Used:

- Flask

3. Data Layer

This layer stores and manages system data.

Functions:

- Store user records
- Store lost item records
- Store found item records
- Retrieve matching data

Database Used:

- MongoDB / MySQL

Architecture Flow

User → Frontend → Backend → Database

Backend → Matching Algorithm → Notification System

4.3 Data Flow Diagram (DFD)

Step 1 : User Registration/Login

Users create accounts and log in securely.



Step 2 : Report Lost or Found Item

Users submit item details such as:

- Item Name
- Description
- Location
- Date
- Image (Optional)



Step 3 : Data Storage

All records are stored in the database.



Step 4 : Matching Process

The system compares descriptions using NLP algorithms.



Step 5 : Similarity Score Generation

The system calculates similarity scores between records.



Step 6 : Notification

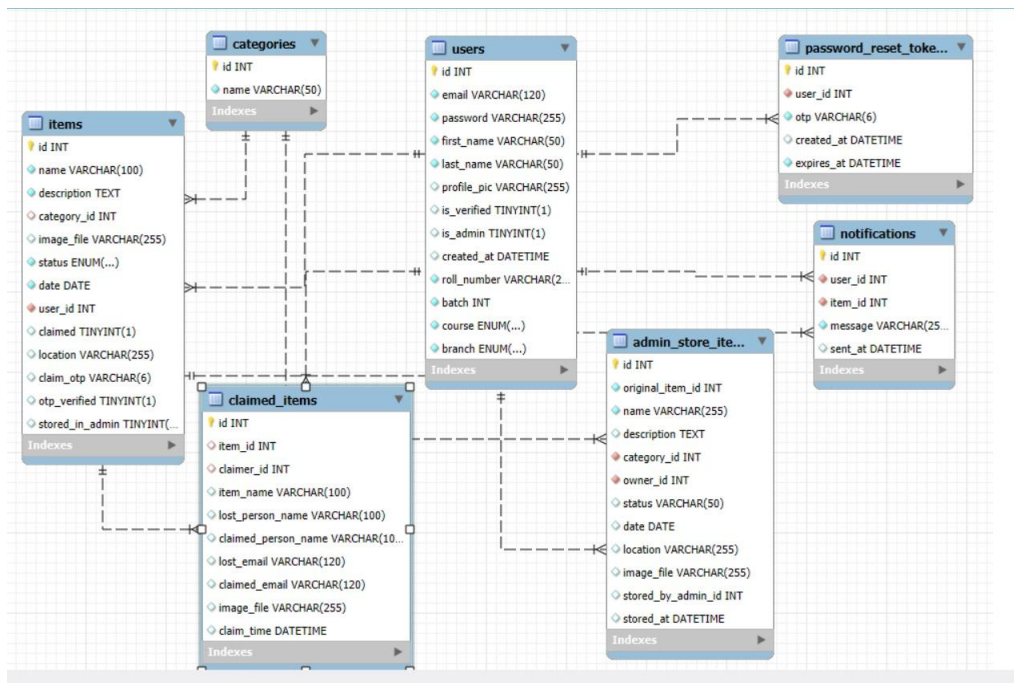
Users receive notifications when a possible match is identified.



Step 7 : Admin Verification

Admin verifies suspicious or fake claims.

4.4 ER Diagram



Class Diagram

User Class

Attributes:

- user_id
- name
- email
- password

Functions:

- register()
- login()

LostItem Class

Attributes:

- lost_id
- item_name
- description
- location
- date

Functions:

- reportLostItem()

FoundItem Class

Attributes:

- found_id
- item_name
- description
- location

Functions:

- reportFoundItem()

MatchingSystem Class

Functions:

- keywordMatching()
- tfidfSimilarity()
- cosineSimilarity()

4.6 System Flowchart

START



User Registration/Login



Submit Lost or Found Item



Store Data in Database



Run Matching Algorithm



Generate Similarity Score



Match Found?



YES → Send Notification
↓
NO → Continue Searching
↓
Admin Verification
↓
END

4.7 Module Design

The system is divided into multiple modules for better organization and scalability.

1. User Module

Functions:

- User Registration
- User Login
- Authentication

2. Lost Item Module

Functions:

- Submit Lost Item Reports
- Update Item Information

3. Found Item Module

Functions:

- Submit Found Item Reports
- Store Found Item Details

4. Matching Module

Functions:

- Keyword Matching
- TF-IDF Calculation
- Cosine Similarity

This is the core intelligent component of the system.

5. Notification Module

Functions:

- Notify Users About Matches
- Display Alerts

6. Admin Module

Functions:

- Verify Claims
- Remove Fake Entries
- Monitor User Activities

4.8 Summary

The system design phase defines the structure and workflow of the Lost & Found System. The proposed architecture combines frontend technologies, backend services, database management, and intelligent NLP-based matching algorithms to provide an efficient and scalable platform.

The design ensures:

- Centralized data management
- Secure communication
- Intelligent matching
- User-friendly operation
- Scalable architecture

The structured module-based design makes the system easy to implement, maintain, and expand in the future.

IMPLEMENTATION

5.1 Introduction

Implementation is the process of converting the system design into a fully functional application. In the Lost & Found System, implementation involves developing the frontend interface, backend server, database integration, and intelligent matching mechanism.

The system is implemented using modern web technologies such as HTML, CSS, JavaScript, Flask and MySQL. The implementation phase focuses on creating a secure, user-friendly, and efficient platform for reporting and recovering lost items.

The complete system is divided into multiple modules such as authentication, item reporting, matching algorithms, notifications, and administrative control.

5.2 Project Structure

The project follows a modular folder structure for better organization and maintenance.

Frontend Folder

Contains:

- HTML Pages
- CSS Files
- JavaScript Files
- Bootstrap Components

Backend Folder

Contains:

- API Routes
- Authentication Logic
- Matching Logic
- Database Connectivity

Database Section

Stores:

- User Records
- Lost Item Reports
- Found Item Reports
- Match Information

5.3 Backend Implementation

The backend is developed using Flask.

The backend handles all server-side operations including:

- User Authentication
- API Management
- Database Communication
- Matching Logic
- Notifications

Major Backend Functionalities

User Authentication

Users can register and log in securely using email and password.

API Endpoints

The backend provides API routes for:

- Registration
- Login
- Reporting Items
- Fetching Matches
- Notifications

Matching Process

The backend compares lost and found descriptions using NLP algorithms.

Database Operations

The backend stores and retrieves information from MySQL.

The backend ensures smooth communication between frontend and database components.

5.4 Frontend Implementation

The frontend provides the user interface of the system.

Technologies used:

- HTML
- CSS
- JavaScript
- Bootstrap

Main Frontend Pages

Registration Page

Allows users to create accounts.

Login Page

Provides secure access to the platform.

Lost Item Form

Users enter details of lost items.

Found Item Form

Users submit found item details.

Dashboard

Displays user reports and matching notifications.

Frontend Features

- Responsive Design
- User-Friendly Interface
- Interactive Forms
- Easy Navigation

The frontend ensures smooth interaction between users and the system.

5.5 Matching System Implementation

The matching mechanism is the core functionality of the project.

The system compares:

- Item Name
- Description
- Location Keywords

Techniques Used

Keyword Matching

Important words are extracted and compared.

Example:

Lost: “Black Wallet”

Found: “Leather Black Wallet”

Common keywords generate a possible match.

TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF converts textual descriptions into numerical vectors.

Benefits:

- Measures word importance
- Improves text comparison
- Reduces irrelevant matching

Cosine Similarity

Cosine similarity calculates similarity between vectors.

Values range from:

- 0 → No similarity
- 1 → Exact match

Higher similarity scores indicate stronger matches.

The system generates similarity scores automatically and identifies possible matches.

5.6 Database Implementation

The system uses MySQL for storing records.

Users Table

Stores:

- User ID
- Name

- Email
- Password

Lost_Items Table

Stores:

- Lost Item ID
- User ID
- Item Name
- Description
- Location
- Date
- Image

Found_Items Table

Stores:

- Found Item ID
- User ID
- Item Name
- Description
- Location
- Image

The database ensures centralized and secure storage of all information.

5.7 Notification System

The notification module informs users whenever a possible match is found.

Notification Process

1. Similarity score is generated.
2. Match exceeds predefined threshold.
3. Notification is sent to users.

Notifications help users recover items faster and reduce manual searching effort.

5.8 Admin Module

The admin module is responsible for managing and monitoring the system.

Admin Functions

- Verify reports
- Remove fake entries
- Manage users
- Monitor activities
- Approve or reject suspicious claims

The admin module improves system security and reliability.

5.9 Error Handling

The system includes error-handling mechanisms to improve reliability.

Errors Handled

Invalid Login

Displays error messages for incorrect credentials.

Empty Fields

Ensures required fields are filled before submission.

Duplicate Reports

Prevents repeated submissions of the same item.

Server Errors

Handles backend and database failures gracefully.

These mechanisms improve user experience and system stability.

5.10 Screenshots

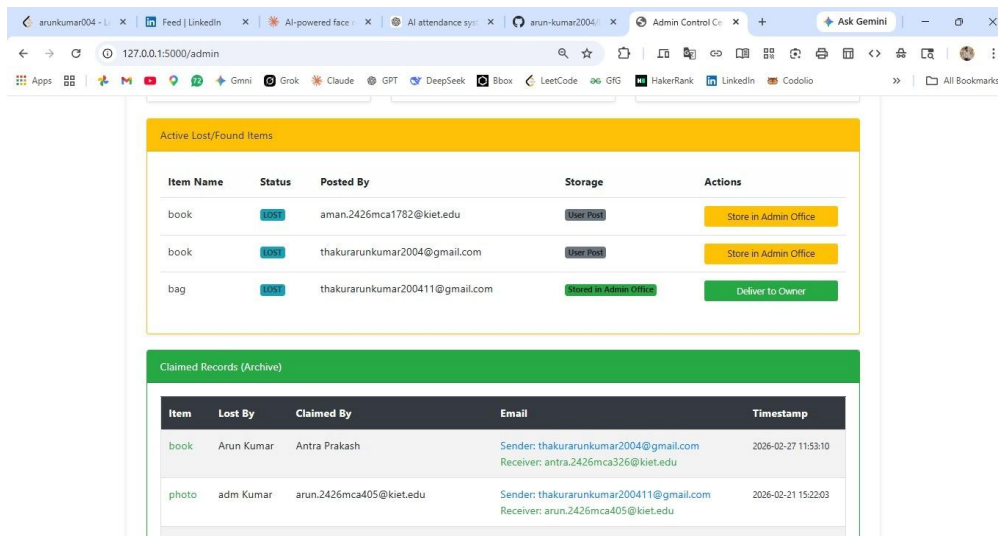
The implemented system includes several interfaces and modules.

Main Screens

- Home Page



• Admin view



The screenshot displays the Admin Control Panel interface. At the top, there's a navigation bar with 'Lost & Found' and 'Home' links, and 'Login' and 'Sign Up' buttons. Below this, a message states 'You have been logged out.' The main content area is divided into two sections: 'Active Lost/Found Items' and 'Claimed Records (Archive)'.

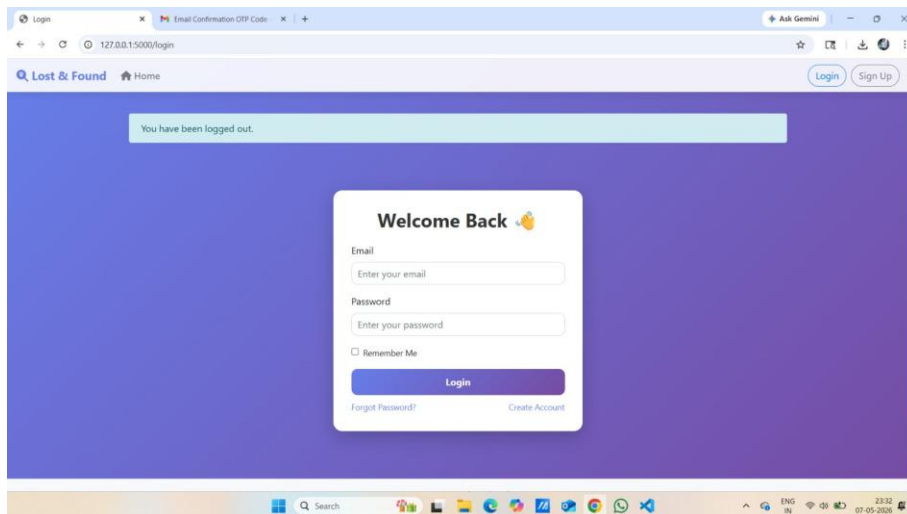
Active Lost/Found Items

Item Name	Status	Posted By	Storage	Actions
book	LOST	aman.2426mca1782@kiet.edu	Used Post	Store in Admin Office
book	LOST	thakurarunkumar2004@gmail.com	Used Post	Store in Admin Office
bag	LOST	thakurarunkumar200411@gmail.com	Stored in Admin Office	Deliver to Owner

Claimed Records (Archive)

Item	Lost By	Claimed By	Email	Timestamp
book	Arun Kumar	Antra Prakash	Sender: thakurarunkumar2004@gmail.com Receiver: antra.2426mca326@kiet.edu	2026-02-27 11:53:10
photo	adm Kumar	arun.2426mca405@kiet.edu	Sender: thakurarunkumar200411@gmail.com Receiver: arun.2426mca405@kiet.edu	2026-02-21 15:22:03

• Login Page



The screenshot shows the Login Page. At the top, there's a navigation bar with 'Lost & Found' and 'Home' links, and 'Login' and 'Sign Up' buttons. Below this, a message states 'You have been logged out.' The main content area features a 'Welcome Back' message with a heart icon. Below the message, there are input fields for 'Email' and 'Password', a 'Remember Me' checkbox, and a 'Login' button. At the bottom of the login form, there are links for 'Forgot Password?' and 'Create Account'.

Welcome Back ❤️

Email
Enter your email

Password
Enter your password

☐ Remember Me

[Forgot Password?](#) [Create Account](#)

[Login](#)

• Registration Page

The screenshot shows a web browser window with the URL 127.0.0.1:5000/register. The page has a purple gradient background. A white form titled 'Create Account' is centered. It contains the following fields: 'Email' (with a hint 'Enter college email' and a note 'Use @kiit.edu email'), 'First Name' and 'Last Name' (split input), 'Roll Number', 'Batch', 'Course' (dropdown menu showing 'Btech'), and 'Branch' (dropdown menu showing 'CSE'). There is a 'Password' field with a hint 'Create strong password' and a note 'Min 8 chars, 1 uppercase, 1 number, 1 symbol'. A blue 'Register' button is at the bottom of the form. The browser's taskbar at the bottom shows various application icons and the system clock indicating 23:32 on 07-05-2026.

- **Lost and Found Items**

The screenshot shows the 'Lost & Found Items' dashboard in a web browser. At the top, there's a navigation bar with 'Lost & Found', 'Home', 'Add Item', and 'Admin' links. A green banner at the top of the main content area says 'Login successful!'. Below this is a search bar with the text 'Lost & Found Items', a 'Search items...' input field, and a dropdown for 'All Categories'. The 'My Posted Items' section displays two items: 'Data Structures Notebook' and 'College ID Card', each with a thumbnail image and a 'View' button. The 'All Items' section is visible at the bottom. The browser's taskbar at the bottom shows the system clock indicating 23:33 on 07-05-2026.

5.11 Summary

The implementation phase successfully integrates frontend development, backend services, database management, and intelligent matching algorithms into a working web-based application.

The system provides:

- Centralized item management
- Intelligent automatic matching
- Notification mechanisms

- Secure user authentication
- Admin verification

The modular implementation ensures scalability, maintainability, and efficient system performance.

TESTING AND RESULTS

6.1 Introduction

Testing is an important phase in software development that ensures the system works correctly and satisfies all project requirements. The purpose of testing in the Lost & Found System is to verify that all modules function properly, matching algorithms generate accurate results, and users can interact with the platform smoothly.

The testing process helps identify errors, improve performance, and ensure system reliability before deployment.

The system was tested under different conditions to evaluate:

- Functionality
- Performance
- Security
- Reliability
- Matching Accuracy

6.2 Testing Methodology

The testing process followed a systematic approach.

Requirement Analysis

System requirements and expected outputs were identified before testing.

Test Planning

Different testing methods and test cases were prepared.

Test Execution

All modules were tested individually and collectively.

Result Analysis

Outputs were compared with expected results.

Error Correction

Detected bugs and issues were fixed and retested.

6.3 Types of Testing

1. Unit Testing

Individual modules were tested separately.

Modules tested:

- Login Module
- Registration Module
- Matching Module
- Notification Module

2. Integration Testing

Integration testing verified communication between different modules.

Example:

- Frontend ↔ Backend
- Backend ↔ Database
- Matching Module ↔ Notification System

3. System Testing

The complete system was tested as a whole to ensure all functionalities worked correctly.

4. User Acceptance Testing

Users tested the platform to verify usability and overall functionality.

Feedback was collected regarding:

- User Interface
- Navigation
- Matching Accuracy
- Ease of Use

6.4 Test Cases

Module	Test Case	Expected Result	Status
Login	Valid Credentials	Login Successful	Pass
Login	Invalid Password	Error Message	Pass
Registration	Empty Fields	Validation Error	Pass
Lost Item Report	Valid Submission	Data Stored	Pass
Found Item Report	Valid Submission	Data Stored	Pass
Matching System	Similar Descriptions	Match Generated	Pass
Notification	Match Found	Notification Sent	Pass

6.5 Testing Environment

Hardware Configuration

Component	Specification
Processor	Intel i5
RAM	8 GB
Storage	256 GB SSD

Software Configuration

Software	Version
Flask	Latest
MySQL	Latest
VS Code	Latest
Chrome Browser	Latest

6.6 Results and Analysis

The testing results showed that the system performed successfully under different conditions.

Functional Results

- User registration worked correctly.
- Login authentication functioned properly.
- Lost and found reports were stored successfully.
- Matching algorithms generated accurate similarity scores.
- Notifications were triggered correctly.

Matching Accuracy

The NLP-based matching mechanism successfully identified similar descriptions even when wording differed.

Example:

Lost Description:

“Black Samsung phone with cracked screen”

Found Description:

“Samsung mobile black color broken display”

Result:

High Similarity Score Generated

This demonstrates the effectiveness of TF-IDF and cosine similarity algorithms.

Performance Results

The system handled multiple reports efficiently and generated results quickly.

Average response time remained low even with increasing records.

Security Results

Authentication and admin verification prevented unauthorized access and fake claims.

6.7 Bug Tracking and Resolution

During testing, several minor issues were identified and corrected.

Bug ID	Issue	Solution
001	Invalid form submission	Added validation
002	Duplicate reports	Added duplicate checking
003	Notification delay	Improved backend processing

After corrections, the system performed reliably.

6.8 Accuracy Analysis

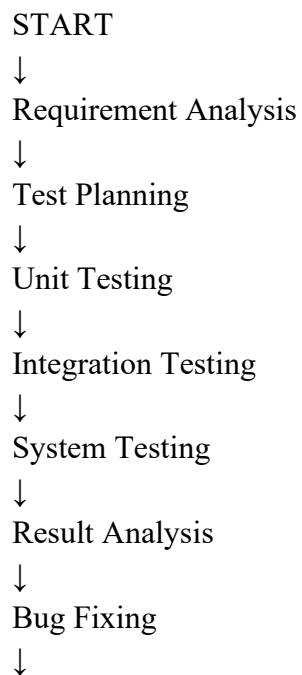
The matching mechanism produced accurate results for most test cases.

Accuracy Benefits

- Faster item recovery
- Reduced manual comparison
- Improved matching efficiency
- Better user experience

The system successfully handled various description formats and generated relevant matches.

6.9 Testing Flowchart



Final Approval



END

6.10 Conclusion of Testing

The testing phase confirmed that the Lost & Found System functions effectively and satisfies all project requirements.

The system demonstrated:

- Reliable performance
- Secure operation
- Accurate matching
- Efficient notification handling
- User-friendly interaction

Overall, the testing results prove that the proposed system is suitable for real-world deployment in colleges, offices, airports, and public environments.

CONCLUSION & FUTURE SCOPE

7.1 Conclusion

The Lost & Found System with Automatic Matching provides an intelligent and efficient solution for managing lost and found items digitally. Traditional methods such as notice boards, handwritten registers, and social media announcements are slow, unorganized, and heavily dependent on manual effort. These systems often fail to establish proper communication between the owner of a lost item and the finder.

The proposed system overcomes these limitations by introducing a centralized web-based platform integrated with intelligent matching algorithms. Users can easily register, log in, and report lost or found items with details such as item name, description, location, and images. All records are securely stored in a centralized database, allowing easy management and retrieval of information.

The major strength of the system lies in its automatic matching mechanism. By using Natural Language Processing techniques such as Keyword Matching, TF-IDF, and Cosine Similarity, the system intelligently compares descriptions and identifies possible matches even when the wording differs. This significantly reduces manual effort and improves recovery success rates.

The system also includes features such as:

- User authentication
- Notification mechanism
- Admin verification
- Centralized database management
- Secure communication

The project demonstrates the practical implementation of web technologies, database systems, and AI-based text similarity techniques in solving a real-world problem. It contributes toward digital transformation and efficient resource management in public and institutional environments.

Overall, the Lost & Found System provides a scalable, secure, and intelligent platform that improves the process of recovering lost belongings while reducing time and administrative workload.

7.2 Future Scope

Although the current version of the Lost & Found System provides efficient functionality, several advanced features can be added in the future to further improve the system.

1. AI-Based Image Recognition

Advanced computer vision techniques can be integrated to compare uploaded images of items and improve matching accuracy.

2. Mobile Application Development

Android and iOS applications can be developed to provide easier access and real-time notifications for users.

3. GPS-Based Location Matching

Location-based services can help identify nearby matches and improve recovery speed.

4. Real-Time Push Notifications

Push notification systems can instantly inform users whenever a possible match is detected.

5. Chat System Between Users

A secure in-app messaging feature can allow direct communication between the owner and finder.

6. QR-Code Based Identification

QR-code tagging systems can help identify and recover registered belongings more efficiently.

7. Cloud Deployment

The system can be deployed on cloud platforms such as AWS or Azure for better scalability and accessibility.

8. Advanced AI Models

Machine Learning and Deep Learning algorithms can further improve text similarity and image recognition performance.

Summary

The Lost & Found System with Automatic Matching successfully combines web technologies, database management, and intelligent NLP-based algorithms to provide an effective solution for lost item recovery. The system improves recovery efficiency, reduces manual effort, and provides secure and centralized management of records.

Future enhancements can transform the project into a large-scale smart recovery platform suitable for educational institutions, transportation systems, offices, airports, and public infrastructures.

REFERENCES

1. MongoDB Documentation
<https://www.mongodb.com/docs>
2. MySQL Documentation
<https://dev.mysql.com/doc>
3. Scikit-learn Documentation
<https://scikit-learn.org>
4. W3Schools Web Development Tutorials
<https://www.w3schools.com>
5. MDN Web Docs
<https://developer.mozilla.org>
6. Visual Studio Code Documentation
<https://code.visualstudio.com>